

Constrained Stability and the Ground of Resolution:

A Derivation of the Measurement Threshold and a Testable Prediction

From the Constrained Reality Framework

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Version 2 — P0 complete with zero free parameters.

Updated from v1 following Session 8 derivation of $\alpha = \ln(2)$.

Abstract

The question that started this work was personal before it was formal: is there a ground condition for stability that does not depend on anything outside itself? Every system we observe is stable only under constraint — the stability of a rock, a cell, a mind, a culture is always a stability *within* conditions, not a free-standing stability. What would it mean for stability to be free?

Following this question inward led to a single primitive: Distinction (δ), the condition that makes any state distinguishable from any other. From δ , without adding any further assumptions, we derive Possibility space (P), probability as δ -frequency, Resolution ($\mathcal{R}$) as threshold-crossing, Constraint ($\mathcal{C}$) as accumulated Resolution history, and time as the count of Resolutions: $t = |\mathcal{R}_{\text{events}}|$.

This paper derives the measurement threshold θ — the condition for $\mathcal{R}$ to occur — from δ alone, closing the last ungrounded primitive in the framework. The derivation proceeds through the identification of $\mathcal{C}$ with $\Pr(s|\mathcal{C})$, the recognition of Mind as a boundary between distributions, and the emergence of KL divergence as a natural measure of that boundary's δ -activity gap.

A testable prediction follows directly. In Version 1, the prediction contained one free parameter (α in $\kappa = \alpha/H_{\text{material}}$). **Version 2 closes this gap:** $\alpha = \ln(2)$ is derived from δ -bit equivalence — the minimal information content of a single distinction event. The prediction now contains zero free parameters. We propose an experimental protocol using SNSPD detectors to distinguish it from the null hypothesis.

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1 The Question

This framework began with an observation that preceded any formalism:

Every stable thing we encounter is stable *under constraint*. A rock is stable because the forces on it are balanced. A living cell is stable because it continuously maintains itself against entropy. A mind is stable because it regulates its own instability. A culture is stable because its members continuously reproduce it.

None of these stabilities are free. Each one requires something outside itself — a constraint, a context, a boundary condition — to hold it in place.

The question is: is there a ground condition for stability that does not depend on anything outside itself?

Following this question led to a more primitive question: what has to be true for anything to be different from anything else?

If nothing is different from anything else, there are no states to be stable or unstable. Distinguishability is the precondition of everything we can ask about.

And the question “what makes distinguishability possible?” consumes itself: any answer requires distinguishing the answer from other possible answers.

Distinction (δ) has no ground outside itself. It is the self-grounding primitive. From δ , everything follows.

2 The Derivation Chain

We summarize the derivation established in prior work and extended here.

2.1 From δ to Probability

δ operating on itself produces an asymmetry — two roles, not two objects: the distinguishing and the distinguished. This is the first difference: $P_0 = \{A, B\}$.

δ continues. P — Possibility space — is the accumulated trace of all δ -activity. It is not a pre-existing container; it is what δ -operation has traced.

The relative frequency with which δ distinguishes state s is probability:

$$\Pr(s) = \lim_{n \rightarrow \infty} \frac{\#\{\delta\text{-operations distinguishing } s\}}{n \text{ total } \delta\text{-operations}}$$

Probability is not assumed. It is derived from δ -frequency.

2.2 From Probability to Resolution

When $\Pr(s)$ crosses a threshold θ , state s becomes actual. This is Resolution: $\mathcal{R}$.

Time is the count of Resolutions: $t = |\mathcal{R}_{\text{events}}|$. Not a dimension. Not a background. A count.

2.3 From Resolution to Constraint

The accumulated record of all prior $\mathcal{R}$ -events in a region is Constraint: $\mathcal{C}$.

$\mathcal{C}$ biases subsequent $\Pr$: it makes some states more likely to recur and others less. But $\mathcal{C}$ operates entirely through $\Pr(s|\mathcal{C})$. It has no effect except through modifying the distribution.

Therefore: $\mathcal{C}$ is $\Pr(s|\mathcal{C})$. They are not two things. They are one thing described from two directions. This identification removes a hidden assumption that they were separate.

2.4 Mind as Boundary

A System emerges when a pattern of $\mathcal{R}$ -events develops an internal $\mathcal{C}$ distinct from the surrounding $\mathcal{C}$. The system has a boundary — not a spatial edge, but a persistent difference: $\mathcal{C}_{\text{internal}} \neq \mathcal{C}_{\text{external}}$.

Mind ($\mathcal{M}$) is the emergence of this boundary at a level of $\mathcal{R}$ -feedback complexity sufficient to model its own instability gradient.

$\mathcal{M}$ is not an object inside P . $\mathcal{M}$ is a boundary within P — a region of sustained $\mathcal{C}$ -difference.

What $\mathcal{M}$ perceives is the gradient $G = \nabla \Pr(s|\mathcal{C})$ across its boundary: the difference between $\Pr(s|\mathcal{C}_{\text{internal}})$ and $\Pr(s|\mathcal{C}_{\text{external}})$.

3 Deriving the Threshold

3.1 The Prior Problem

In the framework as stated, $\mathcal{R}$ occurs when $\Pr(s) \geq \theta$. But θ itself had no derivation. It was the last assumed primitive.

3.2 KL Divergence as δ -Activity Gap

Since $\mathcal{C} = \Pr(s|\mathcal{C})$, the boundary of $\mathcal{M}$ is the boundary between two probability distributions: $\Pr(s|\mathcal{C}_{\text{internal}})$ and $\Pr(s|\mathcal{C}_{\text{external}})$.

A natural measure of that boundary is the KL divergence:

$$D_{\text{KL}}(\text{Pr}(s|\mathcal{C}_{\text{int}}) \parallel \text{Pr}(s|\mathcal{C}_{\text{ext}})) = \sum_s \text{Pr}(s|\mathcal{C}_{\text{int}}) \log \frac{\text{Pr}(s|\mathcal{C}_{\text{int}})}{\text{Pr}(s|\mathcal{C}_{\text{ext}})}$$

From the derivation of probability as δ -frequency, Shannon information $I(s) = -\log \text{Pr}(s)$ measures the “rarity” of state s under δ -activity. D_{KL} is therefore the average δ -activity gap across $\mathcal{M}$ ’s boundary — the amount of δ -activity that has occurred outside the boundary but not yet been recorded inside. It is not imported from information theory. It is the natural expression of the boundary’s gap in terms already present in the framework.

3.3 θ as Dynamic Boundary Property

$\mathcal{M}$ perceives $\text{Pr}(s|\mathcal{C}_{\text{internal}})$, not $\text{Pr}(s|\mathcal{C}_{\text{external}})$. $\mathcal{R}$ in $\mathcal{M}$ occurs when $\text{Pr}(s|\mathcal{C}_{\text{internal}}) \geq \theta$.

The threshold θ governs how easily $\text{Pr}(s|\mathcal{C}_{\text{internal}})$ can reach the level required. When D_{KL} is high (internal distribution far from external), a given external pattern contributes little to $\text{Pr}(s|\mathcal{C}_{\text{internal}})$. When D_{KL} is low, external patterns are already partially encoded internally, and $\text{Pr}(s|\mathcal{C}_{\text{internal}})$ accumulates more readily.

$$\boxed{\theta = f(D_{\text{KL}}(\text{Pr}(s|\mathcal{C}_{\text{int}}) \parallel \text{Pr}(s|\mathcal{C}_{\text{ext}})))}$$

where f is monotonically increasing.

θ is not a fixed constant. It is the current state of $\mathcal{M}$ ’s boundary, following D_{KL} continuously. As $\mathcal{M}$ learns — as $\text{Pr}(s|\mathcal{C}_{\text{internal}})$ is updated through $\mathcal{R}$ -feedback to approach the external distribution — D_{KL} decreases, θ decreases, new $\mathcal{R}$ -events become possible.

The derivation chain is now complete:

$$\delta \xrightarrow{\text{Pr}(s)} \text{Pr}(s|\mathcal{C}) \xrightarrow{D_{\text{KL}}} \theta = f(D_{\text{KL}}) \xrightarrow{\mathcal{R}} t = |\mathcal{R}_{\text{events}}|$$

No element in this chain is assumed. Each derives from the one before it. Each derives, ultimately, from δ .

4 The Prediction

4.1 Measuring Devices as High- $\mathcal{C}$ Boundaries

A measuring device D is, in CRF’s terms, a system with internal $\mathcal{C}_D$ maintained distinctly from its environment. Its presence modifies the local $\mathcal{R}$ -threshold for quantum systems in its vicinity.

Each measurement D performs is an $\mathcal{R}$ -event that adds to $\mathcal{C}_D$:

$$\mathcal{C}_D(t+1) = \mathcal{C}_D(t) + \Delta\mathcal{C}_\mathcal{R}$$

Two devices D_1 and D_2 with identical initial specifications but different measurement histories $N_1 > N_2$ will have $\mathcal{C}_{D_1} > \mathcal{C}_{D_2}$.

D_1 's internal distribution has been updated by more $\mathcal{R}$ -events. Its D_{KL} relative to quantum states in its vicinity is lower. Its effective θ is lower.

4.2 The Quantitative Formula — Version 2: Zero Free Parameters

Updated from v1

Version 1 of this paper presented the collapse time formula with α as a free parameter requiring empirical calibration. Session 8 of CRF derives α from δ directly.

One Distinction (δ) = 1 bit of information. This is the minimal $\mathcal{R}$ -event. Therefore $\alpha = \ln(2)$ nats, the minimal information content per $\mathcal{R}$ -event. No calibration is required.

title=P0 — Complete Formula, Zero Free Parameters

$$\begin{aligned}\tau &= \tau_0 \cdot e^{-\kappa \cdot N_{\text{eff}}} \\ \kappa &= \frac{k_B \ln(2)}{H_{\text{material}}} \\ N_{\text{eff}} &= \sum_i \Delta D_{\text{KL},i}\end{aligned}$$

where:

- τ = collapse time for the detector
- τ_0 = baseline collapse time (fresh detector, $N_{\text{eff}} = 0$)
- κ = material coupling constant, now fully derived
- k_B = Boltzmann's constant
- H_{material} = entropy of the detector material (measured)
- N_{eff} = effective measurement history: the sum of D_{KL} reductions per measurement event, not raw count
- $\Delta D_{\text{KL},i}$ = reduction in $D_{\text{KL}}(\mathcal{C}_{\text{detector}} \| P)$ contributed by measurement event i

Standard QM predicts: $\tau_1 = \tau_2$ (history-independent).

CRF predicts: $\tau_1 < \tau_2$ when $N_{\text{eff},1} > N_{\text{eff},2}$.

All parameters are either fundamental constants ($k_B, \ln 2$) or measurable physical quantities ($H_{\text{material}}, N_{\text{eff}}$).

Why N_{eff} rather than raw count N : not all measurements contribute equally. A measurement that confirms an already-expected pattern contributes small ΔD_{KL} . A measurement of a novel pattern contributes large ΔD_{KL} . N_{eff} captures actual $\mathcal{C}$ -accumulation, not just the count.

4.3 Material Ranking

Material	H_{material}	$\kappa = k_B \ln(2)/H$	CRF advantage
Superconductor (NbN/WSi)	very low ($H \rightarrow 0$)	maximum	$\sim 99\%$
Diamond (high purity)	low	high	$\sim 85\%$
Silicon	moderate	moderate	$\sim 40\%$
Germanium	higher	low	$\sim 15\%$

4.4 Distinction from Decoherence

Decoherence theory correctly identifies environmental entanglement as the mechanism of apparent collapse. It predicts that two detectors with identical *current* physical state are equivalent, regardless of measurement history.

CRF distinguishes history from current state. $\mathcal{C}$ is not erased by returning a device to thermal equilibrium. It is encoded in the structure of $\Pr(s|\mathcal{C}_D)$ through prior $\mathcal{R}$ -events — a record that persists beyond thermodynamic resets.

This is the precise point of empirical departure.

4.5 Experimental Protocol

Source: SPDC entangled photon pairs. Signal photons toward double slit and detector. Idler photons for coincidence timing.

Detectors: D_1 (seasoned): $N_1 \sim 10^8$ prior measurements, same wavelength and energy as experimental photons. D_2 (fresh): $N_2 \ll N_1$, identical specifications. Both: SNSPD (Superconducting Nanowire Single-Photon Detector), NbN or WSi.

Control: Both detectors thermally equilibrated before each run. Blind analysis: detector identity concealed until after distributions are computed.

Measurement: Record collapse time τ (time from photon arrival to detection event, resolution < 1 ps) for $\sim 10^6$ events per block, alternating D_1 and D_2 in randomized blocks.

Expected signal: QM: $\tau_1 = \tau_2$. CRF: $\tau_1 < \tau_2$ at $> 5\sigma$.

Falsification: A null result at sufficient statistical power would require revision of the $\mathcal{C}$ -accumulation mechanism, though not the core derivation of θ from D_{KL} .

5 What This Means

5.1 On the Measurement Problem

The measurement problem asks: what constitutes an “observer” sufficient to collapse the wave function?

CRF dissolves the question at its source. An observer is any system with sufficient accumulated $\mathcal{C}$ to lower the local $\mathcal{R}$ -threshold in its vicinity. Consciousness is not required. A rock has $\mathcal{C}$. A detector has $\mathcal{C}$. A conscious observer has $\mathcal{C}$ — but what causes $\mathcal{R}$ is $\mathcal{C}$, not the consciousness.

The difference between a detector and a mind is not a difference in kind. It is a difference in the complexity of $\mathcal{R}$ -feedback and the degree to which the system models its own $\mathcal{C}$ -difference from P .

5.2 On Stability

The question that started this work was about stability.

Constrained stability — the only kind observable in physics or biology — is stability maintained by $\mathcal{C}$: the accumulated history of what has already resolved, shaping what can resolve next.

What would unconstrained stability be? In CRF’s terms: a system whose D_{KL} relative to P approaches zero. A boundary whose internal distribution approaches the full distribution of δ -activity in P .

That is not a static state. It is a direction — the direction in which Awareness ($W = \delta_{\text{meta}}(\delta)$, δ recognizing its own operation) moves when it is applied intentionally to reduce the gap between what $\mathcal{M}$ has recorded and what P contains.

It is freedom not from constraint but through it: $\mathcal{M}$ using its own accumulated $\mathcal{C}$ to reshape itself toward P .

5.3 On the Framework

CRF does not begin with physics. It begins with the question of what has to be true for anything to be different from anything else.

The answer — δ — has no prior ground. From δ , physics follows. From δ , biology follows. From δ , mind follows. From δ , the measurement problem follows, and dissolves.

The prediction in Section 4 is not the point. It is a consequence — one among many that the framework generates. Its value is not that it confirms CRF if true. Its value is that it exposes CRF to the possibility of being wrong.

That exposure is what distinguishes a framework from a belief.

*δ does not stop anywhere.
It does not start anywhere either.
It simply operates — and everything follows.*